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Version: 1

ID: A3YRXSD74VDV-57553043-395

Global Facilities – Design & Construction Standard - I&C – Public Address (PA) System

ECN Facility:	GLOBAL FACILITIES	ECN Area:	ENGINEERING
Owner(s)	ENGINEERING	Target Audience:	All Facilities
Document Type:	Standard	Document ID (pdf):	A3YRXSD74VDV-57553043-395
Version:	1		
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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

This Design & Construction Standard defines the minimum requirements for design, supply, packing and delivery of Public Address System to Micron greenfield project, including installation, cabling, factory & site testing, and commissioning.

Any deviation from this standard shall be submitted for approval during evaluation. Deviation list format shall refer to Appendix A in this standard document.

2.0 Scope

2.1 Site(s) impacted

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

2.2 Vendor Scope

2.2.1 The scope shall be for the design, manufacturing, factory testing, shipping, installation, testing, commissioning & warranty of Public Address system, as well as selection of the suitable type and number of speakers in each room/area to meet the required ambient sound level.

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2.2.2 The package requirements shall include, but not limited to the following:

- a. All components, hardware and software including monitor panels, mixers, amplifiers, event recorders, tuners, battery & charger, supervisory panels, equipment racks, power supplies, loudspeakers, cable ladders/trays/trunkings/conduits, and power and interface cables, etc necessary to function as a complete, operable and independent system.
- b. Cable glands, termination kits, ferrules, cable lugs and miscellaneous materials including cable tag numbers.
- c. Expendable materials including those required during testing and commissioning (cable ties, protective coverings, fuses, etc).
- d. Selection of the suitable type and number of speakers in each room/area to meet the required ambient sound level based on local code.
- e. Obtaining all necessary statutory authority's approvals required for the works. Following contract award, corrections, revisions or schedule delays relating to any authorities comments shall be the vendor's sole responsibility.
- f. Layout drawings.
- g. System Architecture drawing.
- h. Bill of Quantity complete with all details such as Type, Size, Quantity of all devices.
- i. Cable schedule complete with all details such as Type, Size, Length, quantity of all cables.
- j. Integration to Third Party Systems.
- k. Liaison with other system vendors for necessary interfacing.
- l. System grounding.
- m. Program Software.
- n. As-Built Drawings.
- o. Fire rated seals for floor and wall penetrations.
- p. Weatherproof seals for external penetrations.
- q. Calibrated test equipment and apparatus.
- r. Certificate (Green Mark, Calibration, etc)
- s. Operation and Maintenance Manual.
- t. Inspection and Test Plant (ITP)
- u. Factory Acceptance Test
- v. Site Acceptance Test.
- w. System Training.
- x. Documentation.
- y. Servicing and Maintenance.
- z. Warranty

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- 2.2.3 The vendor shall be deemed familiar with all relevant laws, regulations and construction practices in local country. The vendor shall be solely responsible for obtaining all necessary licenses, permit, qualifications, right to use patent, and other statutory and legal approval for the performance of the works. This shall include completion of any authority certification and the signing off by both the respective approved qualified person.
- 2.2.4 The equipment and software proposed by the vendor shall be currently in manufacturer. No custom products shall be allowed. All products shall be supported by manufacturer for a minimum of ten (10) years from date of acceptance, including spare parts, board repairs and software revisions.
- 2.2.5 The vendor shall maintain sufficient spare parts in local country for a period of 5 years after mechanical completion and submit a priced spare parts list before contract award.
- 2.2.6 The vendor shall assign technically competent person(s) to be responsible for all technical matters. The technically competent person(s) shall provide shop drawings, manage site installation work and certify the completion of the works.
- 2.2.7 The vendor shall provide the necessary qualified personnel, apparatus and instruments to perform testing to ascertain the performance and compliance of the installation. All test apparatus shall be calibrated prior to brought onto site with calibration certificates being submitted for verification.
- 2.2.8 Preventive maintenance and protection or equipment from time of start-up until final acceptance shall remain under the scope of the vendor. Consumable spares are to be available for acceptance until date of final acceptance.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities

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Term/Acronym	Definition / Description
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
PA	Public Address
BOQ	Bill of Quantity
ELV	Extra Low Voltage
ITP	Inspection Test Plan
EHS	Environmental, Health, and Safety
ERT	Emergency Response Team

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	- Develop and maintain System Standard document - Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard - Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	- Review and provide technical feedback on System Standard - Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	- Review and provide technical feedback on System Standard - Incorporate System Standard into Project Engineering, Design, and Construction - Document project deviations, exceptions, and exclusions from System Standard - Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	Use System Standard as technical reference

5.0 Applicable Codes and Standards

- a. Latest Local National Standards or Statutory Regulations take precedence over other requirements unless otherwise stipulated.
- b. IEC 60051 Direct acting indicating, measuring instruments
- c. IEC 60073 Colours of indicating lights and pushbuttons.
- d. IEC 60079-0 Explosive atmospheres – part 0: equipment – general requirements.
- e. IEC 60079-1 Explosive atmospheres – part 1: equipment protection by flameproof enclosures “d”.
- f. IEC 60079-2 Explosive atmospheres – part 2: equipment protection by pressurized enclosure “p”.
- g. IEC 60079-7 Explosive atmospheres – part 7: equipment protection by increased safety “e”.
- h. IEC 60079-11 Explosive atmospheres – part 11: equipment protection by intrinsic safety “i”.
- i. IEC 60079-14 Explosive atmospheres – part 14: electrical installations design, selection and erection.
- j. IEC 60079-15 Electrical apparatus for explosive gas atmospheres – part 15: construction, test and marking of type of protection “n” electrical apparatus.
- k. IEC 60079-25 Electrical apparatus for explosive gas atmospheres – part 25: intrinsically safe systems.
- l. IEC 60079-27 Explosive atmospheres – part 27: fieldbus intrinsically safe concept.
- m. IEC 60146 Semiconductor converters.
- n. IEC 60158 General use for control panels.
- o. IEC 60228 Conductors of insulated cables.
- p. IEC 60269 Low voltage fuses.
- q. IEC 60304 Standard colours for insulation for Low-Frequency cables and wires.
- r. IEC 60331 Test on electric cables under fire conditions (fire resistant).
- s. IEC 60332.1 Test on electric and optical fibre cables under fire conditions (flame retardant).
- t. IEC 60363 Short circuit current evaluation – circuit breakers.
- u. IEC 60364 Electrical installation in buildings.
- v. IEC 60417 Graphical symbols for use on equipment.
- w. IEC 60445 Identification of equipment terminals.
- x. IEC 60502 Cables.
- y. IEC 60529 Degrees of protection provided by enclosures (IP code).
- z. IEC 60654 Series: Operating Conditions for Industrial-Process measurement and control equipment.
- aa. IEC 60664 Insulation coordination within LV systems.
- bb. IEC 60801 EMI and RFI immunity.
- cc. IEC 61000 Electromagnetic compatibility (EMC).

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dd.IEC 61285	Industrial-Process Control – Safety of analyzer houses.
ee.IEC 61131	programmable logic controllers.
ff. IEC 61508	All parts – functional safety of electrical/electronic/ programmable electronic safety-related systems.
gg.IEC 61511	All parts – functional safety – safety instrumented systems for the process industry sector.
hh.ISA 5.1	Instrumentation symbols and identification
ii. ISA 5.4	Instrument loop diagrams.
jj. ISA 5.5	Graphic symbols for process displays.
kk.ISA RP12.06.01	Recommended practice for wiring methods for hazardous.
ll. ISA 18.1	Annunciator sequences and specifications.
mm.ISA 18.2	Management of alarm systems for the process industries.
nn. ANSI/NFPA 75	Protection of electronic computer/data processing equipment.
oo.NFPA 72	National Fire Alarm and Signaling Code.
pp.NFPA 318	Protection of Semiconductor Fabrication Facilities.
qq.ICC IFC-2012	International Fire Code.

6.0 System Requirement

- 6.1 The PA system shall conform to the requirements of local code. Audible systems shall provide coverage for indoor and outdoor locations.
- 6.2 PA system shall be powered from designated UPS. The system shall also incorporate an integral battery and charger system providing a back-up period 2 hours at full load.
- 6.3 The PA system shall also be capable to store pre-recorded messages. These messages should be able to be announced in respective zones depending on the input received from external source e.g. fire alarm.
- 6.4 All control systems and workstations shall be time synchronized through the plant master clock system to ensure a correct time discrimination between faults, alarms & events. The time synchronization will be done continuously on the network.
- 6.5 System shall include anti-feedback circuitry to prevent feedback during paging.
- 6.6 Loudspeaker circuit shall be provided with approved means to disconnect the circuit automatically from the amplifier in the event of short circuit and cause a visual indication at the control console to indicate the defective circuit.
- 6.7 The intent of the main equipment is to provide power amplification, equalization, gain management, microphone, and auxiliary level mixing capabilities to provide adequate sound reinforcement

throughout the facility. The system requirements shall include but not limited to the following:

1. Central Processor Modules
 2. Booster Amplifier
 3. Master/Slave Call Stations.
 4. Line supervisory and amplification fail monitoring systems.
 5. 19" equipment rack: the control/amplifier rack shall be based on extendible 19" system.
 6. Horn speakers, ceiling speakers, clean room under raised floor horn speakers, Ex-proof speakers, surface mount speakers.
 7. Digital Message recording / playback module.
 8. Volume control units.
 9. Standby battery charger c/w batteries for 2 hours.
 10. Cabling/conduit/tray
- 6.8 The loudspeakers shall be wired up into separate zones and connected to the central control / amplifier rack. The main PA rack together with the master call station shall be installed in the Fire Command Center.
- 6.9 PA system allow announcement calls in individual zones, building or site wide and shall be a part of the plant emergency evacuation plan.
- 6.10 Every item of equipment used within the PA system, be it apparatus or field devices shall be identified by unique tag number affixed to the apparatus.
- 6.11 The normal operating service conditions have been defined on the data sheets. The system and associated components shall be designed for a minimum life expectancy of 20 years under the specified operating conditions.
- 6.12 The panels shall be afforded a minimum environmental of IP42 for indoor units and IP55 for wet areas or outdoor applications in accordance with IEC 60529.

7.0 Technical Requirement

- 7.1 Loudspeaker shall be located such that they meet the necessary requirements.
- 7.2 In general, one ceiling speaker shall be provided for every 25 square meters in each room such as offices and corridors, while a minimum of one ceiling speaker shall be provided for areas less than 25 square meters such as booths, pantry and toilets. Horn speakers shall be provided for all plant rooms, generator rooms and outdoor areas with high ambient noise.
- 7.3 The PA system shall have the capability to transmit different audio alarms (for Fire Alarm, Gas Leak Alarm, and Chemical Leak Alarm etc) through respective loudspeakers. Adequate amplification shall be provided.

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- 7.4 The loudspeaker line supervision system continuously supervises each loudspeaker line for open, short circuit and earth fault. When a fault is detected, a buzzer shall sound off and the corresponding line fault indicator illuminated.
- 7.5 The amplifier system shall feature input sockets for microphones, sounding recording / reproducing equipment, alarm (gas and fire warning) and other auxiliary devices.
- 7.6 The PA system shall provide a general fault to FMCS.
- 7.7 System shall be designed to cater for input switching. Switching of the zones shall be done on the input side, with each zone having its own amplifier.
- 7.8 When the zones are selected for public addressing, a chime shall first be heard, followed by the announcement. The system shall have a range of tones such that it shall be possible to program different tones for different call stations.
- 7.9 All control and switching equipment shall be centralized in the FCC and equipment rooms.

8.0 Cable Requirement

- 8.1 The system shall use Fire resistant cables in accordance with local code. The overall installation shall be in accordance with local code.
- 8.2 All cores shall be stranded copper conductors and shall be delivered to site with the manufacturer's label intact. All wiring from the main equipment rack in the FCC to the various loudspeaker points shall be carried out using minimum of 1 mm², colour coded twisted pair cable in GI conduit.
- 8.3 The vendor shall prepare a cable schedule, to include cable sizes, cable types, routing, source, destination, cable length, gland requirement and cable tags. The schedules shall be formatted on an excel spreadsheet and extended to cover all cables and building wire installed.
- 8.4 No cable joints or repairs shall be allowed in cables unless specifically approved in writing. Where necessary, the vendor shall arrange with the manufacturer to supply oversize drums to ensure cable lengths are adequate without the need for intermediate joints.
- 8.5 It shall be the responsibility of the vendor to determine the cable lengths and verify the cable size. Voltage drop calculation shall be submitted to verify the selection.
- 8.6 Each indoor loudspeaker shall be terminated through a 2A three-pin plug and socket complying with the latest edition of BS546 or its international equivalent to facilitate easy removal of the loudspeaker.
- 8.7 Outdoor speakers shall be terminated through weatherproof type plug and sockets with a push-on cap and cap retaining ring.

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- 8.8 Where cable glands enter equipment and secured by locknuts, a minimum of two threads shall protrude through the locknut.
- 8.9 All cable glands shall be the nickel coated brass type.
- 8.10 Approved two-hour fire rated seals shall be installed where the cables pass through walls or floors.
- 8.11 With the exception of special cables, the final cable connection shall be via a compression type terminal lug, crimped in one or two positions depending on size using a hand operated mechanical crimping tool.
- 8.12 Sample board shall be submitted showing all the types and sizes of cable specified in cable schedule and type test reports for all the types of cables.

9.0 Cable Tray, Ladder and Cable Trunking Installation.

- 9.1 Splice plated suitable for 15-mm telescopic adjustment shall be provided for joining lengths of cable tray. Splices shall be located within 300 mm of a cable tray support and be secured to the cable tray using bolts, washers and anti-vibration locking nuts. Splice plates shall be shaped to provide a smooth surface at all joints.
- 9.2 Cable trays located outdoor shall be provided with tray covers. Where cable tray or ladder cover is required, they shall be manufactured from sheet metal with a minimum thickness of 1 mm and be the same length as the cable tray or ladder. Covers shall be fixed to the cable tray or ladder using approved metal thread screws located not to damage cable insulation.
- 9.3 The complete installation shall be true and straight, free from twists warps or damaged components. They shall run parallel with the main lines of the building unless otherwise specified. Any support or section of cable tray or ladder poorly installed or of inferior quality shall be removed from the site and new items installed by the vendor at no additional cost.
- 9.4 Cable trays or ladders shall be labelled at 15-meter centers.
- 9.5 Supports shall be approved by Micron before prior to installation.
- 9.6 If cable support is shared with other packages, Public Address system vendor shall provide the cable support divider.
- 9.7 Whenever Cable trunking is used, the inner radius of any bend shall not be less than 2.5 times the minor dimension of rectangular section trunking.
- 9.8 Suitable brackets fabricated from hot dipped galvanized mild steel angle iron shall support trunking at regular intervals not exceeding 1.2 meter.
- 9.9 Vertical runs of trunking within electrical risers that pass through floors must be fitted with 2-hour fire resistant barriers in and around the trunking. These barriers should consist of fire rated material cut away to enable the cables to pass through. Similar fire barriers shall be provided for cable trunking passing through fire rated walls.
- 9.10 Trunking runs shall be completely erected prior to any cable being drawn in. The number of cables installed shall comply with the space factors dictated in the appropriate local code.

10.0 Testing and Commissioning

10.1 General Requirement

- i. Systematic process of ensuring systems perform interactively per Project design intent and Owner's performance requirements.
- ii. The system is to be tested in accordance with manufacturer's recommendations and NFPA or equivalent.
- iii. Inspection and Test Plan should be submitted together with Quality management system for user's approval during design engineering stage.
- iv. The organizing, coordination of personnel, and scheduling of all tests to be the responsibility of the Vendor. Notification of testing to be submitted to each testing participants a minimum of 2 weeks prior to the requested testing date.
- v. Test programs and procedures to be created by the Vendor. Test procedures to have signoff spaces for the Owner and the Vendor. Where Owner-furnished test procedures exist, they are to be used in place of Vendor's test procedures.
- vi. Only qualified personnel shall perform the commissioning of the complete system and Owner acceptance tests.
- vii. Pretest the system for satisfactory operation prior to Owner system acceptance testing. Prior to the Owner witnessing any testing, a system printout demonstrating satisfactory pretest results to be submitted to the Owner 2 days prior to the anticipated test date.

10.2 FAT (Factory Acceptance Test)

- i. All factory tests shall be performed in accordance with EIA RS-276 and RS-490
- ii. The equipment will be inspected during manufacturing and before shipment for site installation. Furthermore, Micron reserves the right to perform formal shop inspections. Vendor shall advise Micron of the fabrication status on a monthly basis. Prerequisite for shipping is the Micron approved FAT.

10.3 Installation Inspection

10.3.1 The equipment and panel pass a visual inspection and complies with the following:

- i. Outer enclosure is labeled appropriately with ID tag.
- ii. Devices and modules are installed per the as-built drawings.
- iii. Wiring is gathered with tie-wraps or bundled within wire-mold channels.
- iv. Wiring, devices and modules are labeled within the panel in accordance with the drawings.
- v. Wiring terminations are adequate and firm.
- vi. Wiring cable shields are terminated at one end to power supply ground as appropriate.
- vii. Confirm panel and equipment rack are grounded to appropriate grounding source.

10.3.2 Devices are mechanically fastened properly, securely and firmly on mounting supports.

10.3.3 Field cable installation shall be visually inspected as per cable schedule.

10.4 SAT (Site Acceptance Test)

10.4.1 The final acceptance at site will be carried out after mechanical completion of the system at Micron facility. The purpose of the SAT is to verify that the system is supplied and installed and that it operates according to the relevant specification.

10.4.2 At the completion of installation of panels and field devices, each system to be tested by the Vendor with a factory-trained and certified field technician to show the following:

- i. Panel Verification Testing shall be at least, but not limited to, the following:
 - Jumper and dip switch settings are in accordance with the manufacturer's requirements.
 - Confirm input voltage is within appropriate tolerance range, and that the panel indicates normal operating condition upon the power application.
 - Confirm power supply output voltages are within appropriate tolerance range and that the battery and charger indicate proper charge and operation where batteries and battery charging systems are used.
 - Simulate incoming power failure and test all components when system is operating on battery. Confirm no loss of functionality occurs.
 - The communication and device operation within each rack, between equipment and head to head controls as an entire system is established and verified.

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- The appropriate communication and alarm announcement are demonstrated upon a test simulation executed.
 - Defects to be corrected at once and the test re-conducted at no additional cost to the Owner until the defect as been corrected.
- ii. Site testing shall also include wiring loop checks and simulation tests.
 - iii. Test on coverage and intelligibility of voice announcements. Tests shall be conducted from the master call station. The methods used for the measurement of sound pressure levels and intelligibility of speech and sound shall conform to the requirements of relevant IEC/BS and ANSI S1.13 and S3.2.
 - iv. Test for system frequency response from call station to loudspeaker output and pre-amp to loudspeaker output at SPL.
 - v. Site test shall include dispersion of loudspeaker between 80Hz to 20kHz, cable isolation from shields and ground, and test on the CD/recorded player and radio tuner.
 - vi. The manufacturer shall provide certificates for all tests. The test certificates shall include values of the test results

10.5 Documentation

- 10.5.1 Quality Management System should be in place and used for documentation of the commissioning activities.
- 10.5.2 The Vendor should provide a document list of his standard documentation and commissioning documentation package together with examples for review and final approval by Micron including a description of the vendor internal quality system.
- 10.5.3 Prior to the execution of any commissioning activity of the vendor, Micron is to be supplied with the respective protocols and test specification for approval – these must be made available to Micron at least 4 weeks before execution of the testing.

11.0 Document Control

11.1 **Approval:** GLOBAL_FAC_GFTT_APPR

11.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10W_FAC_NOTIFY
FCG_F10N_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTTW_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
MTB_FAC_ALL_NOTIFY

11.3 Final Viewer Access

All Micron Team Members at all Sites

11.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#)

11.5 Review

Biennially (two years)

12.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	10/30/20

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Appendix A: Deviation List

Any deviation from this specification shall be submitted for approval during vendor evaluation. Deviation list format shall be the following:

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT